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A three-dimensional liquid diode for soft integrated permeable electronics

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Wearable electronics with great breathability enable a comfortable wearing experience and facilitate continuous biosignal monitoring over extended periods¹⁻³. However, current research on permeable electronics is predominantly at the stage of electrode development, which is far behind the practical applications with comprehensive integration with diverse electronic components (e.g., circuitry, electronics, encapsulation)⁴⁻⁸. Achieving permeability and multifunctionality in a singular, integrated wearable electronic system remains a formidable challenge. Here, we present a general strategy for integrated moisture-permeable wearable electronics based on three-dimensional liquid diode (3D LD) configurations. By constructing spatially heterogeneous wettability, the 3D LD unidirectionally self-pumps the sweat from the skin to the outlet at a maximum flow rate of 11.6 ml/cm²/min, 4000 times greater than the physiological sweat rate during exercise, presenting exceptional skin-friendliness, user comfort and stable signal reading behavior even under sweating conditions. A detachable design incorporating a replaceable vapor/sweat-discharging substrate enables the reuse of soft circuitry/electronics, increasing its sustainability and cost-effectiveness. We demonstrated this fundamental technology in both advanced skin-integrated electronics and textile-integrated electronics, highlighting its potential for scalable, user-friendly wearable devices.

Continuous monitoring of vital/physiological signals over extended temporal windows plays a significant role in a comprehensive perspective on an individual's overall health status, enabling early disease prediction, self-directed diagnostics, personalized therapeutics, and improved management of chronic health conditions⁹⁻¹¹. Recent advances in wearable electronics, particularly those incorporated within skin-integrated or textile-based ones, have facilitated continuous on-body biosignal monitoring during daily activities¹²⁻¹⁵. However, despite significant progress in optimizing the mechanical and electrical properties of these wearable devices, several barriers to their widespread adoption and clinical applications remain¹⁶⁻¹⁸. One major challenge is the poor permeability of electronic materials/devices, leading to sweat accumulation at the skin-device interface after a certain time of operation^{2,19}. This hurdle not only causes thermophysiological discomfort to users, but also deteriorates collected signal quality and adhesion property, thereby hindering precise signal monitoring over prolonged periods²⁰.

Breathable electronics based on ultra-thin^{7,21}, porous^{5,22}, nanofiber^{6,23}, nanomesh^{4,24} or textile-based architectures^{8,25}, enable the passive or active discharge of gases, vapors and sweat through their micro/nano openings. Thereby, they offer a user-friendly interface between the skin and the device, ensuring the possibility for more stable signal acquisition during long-term healthcare monitoring even under sweating conditions²⁶⁻²⁸. However, from the standpoint of practical applications, the development of breathable electronics is still in its infancy stage, as the functions of these devices are mainly associated with simple electrodes²⁹, sensors³⁰, antennas²², energy harvesters/storage^{31,32}, and displays⁸ (Supplementary Fig. 1 and Supplementary Table 1). The state-of-the-art wearable electronics include much more

1 complicated functional sensors, circuits, modules for signal acquisition, processing,
2 transmission¹¹. Therefore, it's still a grand challenge to realize high-level integrated and multi-
3 functional wearable electronics yet with a breathable format³³⁻³⁶.

4 Here, we report a fundamental methodology from material processing, device architecture,
5 and system integration for integrated permeable wearable electronics based on a concept of 3D
6 LD (Fig. 1a), which can serve as a universal platform/substrate to integrate with the well-
7 established skin-integrated and wearable electronics. Distinguished from the reported works,
8 the 3D LD doesn't rely on unique materials but adopts an in-plane liquid transport layer called
9 horizontal liquid diode (HLD) on top of the out-of-plane perspiration channels called vertical
10 liquid diode (VLD). Combining in-plane and out-of-plane sweat transportation features, the 3D
11 LD not only offers great air/sweat permeability but also allows high-performance wearable
12 electronics to be directly integrated on top of it with no effect on perspiration (Fig. 1b and
13 Supplementary Note 1). The 3D LD with nature-inspired microstructures allows directionally
14 and spontaneously transport fluids to continuously transport sweat in a pre-programmed
15 path^{37,38}. It can rapidly self-pump sweat from the skin-device interface to the outlet, yet prevent
16 backflow, which enables the wearable device to provide reliable biosignal monitoring, robust
17 adhesion strength, and great comfortability with irritation-free during long-term wear (Fig. 1c-
18 e). Furthermore, the detachable design based on the magnetic coupling of 3D LD substrate and
19 flexible/wearable electronics significantly enhances the practical usability and reduces costs
20 (Fig. 1f).

21 **Results**

22 **Design of the permeable integrated system**

The idea of integrated sweat-permeable electronics is illustrated in Figs. 1b and 2a, wherein orderly integration of serpentine metallic interconnects serves as permeable electrodes, 3D LD both serves as a substrate for directional sweat expulsion and supports multifunctional circuits/electronics with technical mature processing methods. Parameters of human sweat ducts are approximately associated with a diameter of 100 μm and a minimum interval of 400 μm (Supplementary Fig. 2)^{26,39}. We employed serpentine meshed interconnects of polyimide (PI)/Gold (Au) ribbons with a coverage ratio of 30 % as the electrode (Fig. 2b). This design not only offers conformal contact with the skin for maintaining low skin-electrode impedance and stable electrical conductivity under deformation, but also exhibits superior permeability of the electrode (Fig. 2c, d and Supplementary Figs. 3-5). The VLD featuring with a hydrophilicity gradient channel, can rapidly pump sweat generated at the skin-device interface to the back side (Fig. 2e, Supplementary Fig. 6a and Note 2). The spatially distributed design of VLD channels enables sweat to form as droplets rather than disperse onto the surface, unlike the conventional Janus fabric, which is beneficial for subsequent detachment (Fig. 2f). To guarantee unobstructed sweat discharge toward the outlet, we also introduced a HLD with in-plane liquid transport ability (Fig. 3a, Supplementary Fig 6b and Note 2). The HLD allows sweat to spread inside the channel with a default pathway to enable it to ultimately reach the sweat collector, where it either evaporates into the surrounding environment or flows through the outlet and subsequently drips. Since all the pores for dripping sweat are designed at the edge of the 3D LD substrate, the wearable electronics can be directly integrated into the middle of the substrate without impeding the perspiration pathway (Fig. 1b).

Characterizations of the VLD

1 The development of the VLD starts from the superhydrophobic treatment of a polyester
2 fabric (150 μm thick; Supplementary Figs. 7, 8). Single surface dip-coating of a water-based
3 pressure-sensitive adhesive (PSA) enables fabric exhibiting strong adhesion while maintaining
4 its porous structure (Supplementary Figs. 9-12). Selective plasma treatment is employed to
5 generate a wettability gradient throughout the exposed channel vertical direction (Extended
6 Data Fig. 1a). The exposed area (plasma-treated region) and plasma treatment duration are
7 significantly associated with the performance of the VLD. We first measured the contact angles
8 of a single exposed channel under varying treatment conditions. The results show that fabrics
9 with larger exposed areas presented directional sweat transport abilities at shorter treatment
10 duration, while longer treatment duration breaks through the liquid diode (Extended Data Fig.
11 1b). A needle featuring controllable flow was employed to evaluate the VLD's sweat transport
12 capabilities under different treatment conditions (Fig. 2g and Supplementary Fig. 13 and Video
13 1). A trade-off between the exposed area and treatment duration affects the sweat transport rate,
14 where the maximum flow rate of a single channel can reach 1.4 ml/min (mask size of 1.5 mm,
15 treatment duration of 0.5 min; Fig. 2h).

16 In addition to the sweat transport rate, the final size of the hydrophilic area (physically
17 treated region, due to the plasma diffraction) is another crucial factor influencing the VLD's
18 performance, the channel arrangement, density, and surface tension. Figure 2i illustrates the
19 dependence of hydrophilic area on treatment conditions, indicating that the practical area
20 typically exceeds the exposed area. This phenomenon arises from plasma diffraction/diffusion
21 within the channel along the porous fabric, as confirmed by surface tension measurements
22 (Supplementary Fig. 14). Despite the peak transport rate of a single channel at a mask size of

1 1.5 mm (1.4 ml/min), the extensive hydrophilic area (diameter of 1.76 mm) complicates the
2 close arrangement of the channels. Conversely, a mask size of 0.2 mm results in the smallest
3 hydrophilic area (diameter of 0.87 mm), but the low sweat transport rate (0.3 ml/min) limits its
4 efficacy in sweat discharge. Consequently, the optimal design parameters were determined to
5 be a mask size of 0.4 mm and a duration of 3 minutes, which presented a high sweat transport
6 rate (0.9 ml/min) and a small hydrophilic area (diameter of 0.92 mm) simultaneously. Further
7 parameters, such as fabric thickness, dip-coating repetitions, and superhydrophobic treatment
8 duration, were also studied for their influence on the performance of the VLD (Supplementary
9 Figs 15-17 and Note 3). The channel density arrangement was corroborated by the sweat
10 cohesion test of adjacent channels (Supplementary Fig. 18), which revealed that sweat droplets
11 are more readily to connect at neighboring channels with small intervals. The results show that
12 the VLD, with a 3 mm interval, concurrently presents a high channel array density and a low
13 droplet cohesion trend, thus proving to be the optimized parameter for the VLD. With this
14 optimal design, the VLD shows a high sweat transport rate of 11.6 ml/cm²/min, which is three
15 orders of magnitude superior to the human sweat rate during mild exercise ($0.38\text{--}2.85\times 10^{-3}$
16 ml/cm²/min; Fig. 2j and Supplementary Fig. 19)⁴⁰.

17 Furthermore, we assessed two key performances of the VLD, namely breathability (water
18 vapor transmission rate, WVTR) and stability (both mechanically and long-term operationally).
19 In comparison to the pristine fabric, both superhydrophobic fabric and the VLD exhibit
20 negligible differences in WVTR (Fig. 2k). The VLD also showcases consistent sweat transport
21 capabilities under varying deformations and different batches (Fig. 2l and Supplementary Figs.
22 20, 21). In the stability test over 30 days, artificial sweat at 0.6 ml/cm²/day, mimicking football

1 players' daily sweat loss⁴¹, passing through the VLD. The device consistently showed a high
2 transport rate of 8.1 ml/cm²/min even after one month (Fig. 2m). Note, this value is still far
3 greater than the typical sweating rate of humans during physical exercise⁴⁰.

4 **Characterizations of the HLD**

5 The unidirectional sweat transport capability of the HLD is attributed to the meticulously
6 engineered gradient hydrophilic micropillars with a careful mechanics design (Fig. 3a). A
7 coating of superhydrophilic material, polyvinyl alcohol (PVA) and silica nanoparticles
8 composite on the polydimethylsiloxane (PDMS) micropillars enables long-term durable
9 hydrophilicity to the surface (Fig. 3b and Supplementary Fig. 22). These spatially distributed
10 micropillars exhibit uniform dimensions and height, while cross-shaped supports safeguard the
11 micropillars against damage under vertical pressure (Fig. 3b, c). Finite element analysis (FEA)
12 modeling corroborates the safeguarding role of the cross-shaped supports, where it can be found
13 stress predominantly concentrates at the supporting structures when force is applied (Fig. 3c
14 and Supplementary Fig. 23). The support parameters are ascertained based on an assessment of
15 permeability and mechanical properties. It reveals that breathability of the HLD increases with
16 the supporting structure's height, achieving its maximum at a 4 mm interval (Supplementary
17 Fig. 24). FEA findings reveal a significant increase in film deflection under gravitational
18 influence when the interval exceeds 4 mm, suggesting insufficient support for the film
19 (Supplementary Fig. 25). Despite the 2 mm interval support offering superior mechanical
20 properties, the diminished permeable space substantially impedes the HLD's breathability. The
21 experimental and theoretical findings corroborated that the optimal interval is 4 mm. The
22 micropillar arrangement significantly impacts the sweat dynamics of the HLD, and we found

the distance between adjacent rows of pillars to be the critical factor (Supplementary Fig. 26). Experimental evidence demonstrates that the HLD's transport rate diminishes as the interval between rows increases, while the pinning breakdown position exhibits an inverse trend (Supplementary Figs. 27-31). Taking this trade-off into account, an optimized parameter of 100 μm for the rows' interval is established, resulting in both a high transport rate and an extensive breakdown distance.

PDMS is recognized as an advanced material for wearables and microfluidics, but its inherent hydrophobicity prevents its adoption in self-pumping liquid transport systems (Fig. 3d). We employed a composite coating (PVA/SiO₂) for extended surface superhydrophilic treatment of the PDMS (Supplementary Fig. 32). The PVA/SiO₂-coated PDMS surface showed superhydrophilicity over a 30-day testing period (Fig. 3e). We found that the contact angle of the surface would significantly affect the transport rate of the HLD, where smaller contact angles lead to increased transport rates (Supplementary Figs. 33, 34). The transport rate of the HLD with a superhydrophilic coating can significantly increase upon surface moisturization by sweat (Fig. 3f, g and Supplementary Fig. 35 and Video 2). Another determinant of the transport rate is the height of the micropillars, where the transport rate increases as the micropillar height increases (Supplementary Fig. 36). However, a trade-off exists between the transport rate and structural stability. The height of the micropillars > 100 μm would cause broken pillars and bent morphology due to the high aspect ratio (Supplementary Fig. 37). Moreover, pillars with a high aspect ratio also exhibit structural instability under sweat flow flushing, and the experimental results are consistent with the FEA (Supplementary Figs. 38-42). Ultimately, the diameter of the micropillars also emerges as a crucial factor to be optimized for amplifying the

sweat-discharge process efficiency, which modulates the dynamic flow within the HLD, especially in terms of fluid velocity and discharge flow rates (Supplementary Fig. 43). The flow velocity amplifies with the enhancement of the pillar diameter while the flow rate peaks at a diameter of 50 μm . With optimized parameters (50 μm diameter, 100 μm height), the HLD demonstrates stable transport capability under deformation and prolonged sweat flushing (Supplementary Figs. 44, 45).

Characterizations of the 3D LD

The 3D LD integrated with VLD and HLD can be realized by inserting an annular hydrophilic PET fabric, as which serves as both the adhesive layer and perspiration collector (Supplementary Fig. 46). The 3D LD shows a superior capability in the transportation of sweat from the skin to the designated outlets (Supplementary Fig. 47). FEA modeling of velocity magnitude, flow streamlines, pressure contours and spatiotemporal volume fraction also proves the high efficiency of the 3D LD (Extended Data Fig. 2 and Supplementary Video 3). Fig. 3h shows the comparison of the perspiration performance between the 3D LD and most used PDMS substrate, where a simplified artificial sweating model simulates the perspiration process (Supplementary Video 4). This model incorporates an internal chamber with a perforated surface and is connected to a syringe pump with a flow rate of 0.1 $\text{ml}/\text{cm}^2/\text{min}$ (nearly 40 times of human sweat rate during physical exercise)⁴⁰. We positioned the PDMS thin film substrate and the 3D LD on top of the perspiring pores to examine fluid dynamics under continuous sweat flow. The continuous liquid injection heightened interfacial pressure, leading to the formation of a dome shape in the PDMS film due to its low gas and water permeability. In this scenario, further liquid injection caused the fluid to breach the interface between the PDMS and artificial

skin, resulting in a sweat leak and thus a delamination. Conversely, the 3D LD presents superior permeability, permitting unhindered transportation of gas and sweat within the internal channels to the outlets, thus maintaining conformal contact with the artificial skin. Both experimental and FEA outcomes reveal that the 3D LD sustains stable sweat transportation capabilities under various deformations (Fig. 3i, j). Furthermore, cytotoxicity assessments indicate that the 3D LD exhibits great biocompatibility, with no discernible differences between the samples and the control group (Extended Data Fig. 3).

User study of the 3D LD

In the user study, we conducted comprehensive assessments to investigate the impact of sweating/moisture management of different wearable devices/substrates on both wearing comfortability and device performance. The assessments include adhesion tests, evaluations of skin thermal comfort, and comparison of device collected signal quality. To carry out adhesion evaluation, we affixed four different types of patches, including commercial electrocardiogram (ECG) patches, PDMS film, VLD, and 3D LD, to the testing subjects. The subjects then engaged in various physical activities, while we captured images of the patch status (Supplementary Fig. 48). Over the 30-minute test period, we observed that the commercial ECG and PDMS patches detached from the skin during the test, while the VLD and 3D LD patches remained firmly attached to the body (Fig. 4a). We found that the patches' adhesion strength under sweating conditions exhibiting a positive correlation with the permeability and softness of the device (Fig. 4b).

Next, we conducted a long-term wearing test, in which we affixed different patches to the forearms of three individual subjects for three days of continuous use. We observed no irritation

or inflammation for the skin wearing the VLD and 3D LD patches, whereas the commercial ECG and PDMS patches caused obvious skin erythema (Fig. 4c). Furthermore, we applied ten 3D LD patches to various parts of a volunteer's body and no irritation was observed during the 12-hour wear period (Supplementary Fig. 49). We carried out another user test to assess the impact of permeability on user wearable comfortability. We affixed the four types of patches to the forearms of 10 subjects (5 male and 5 female) and instructed them to wear the patches for 12 hours. The subjects were allowed to remove the patches if they experienced any discomfort, and we recorded the duration that they wore each patch. The results indicated that the VLD and 3D LD last the longest wearing durations compared to the others (Fig. 4d). The excellent vapor and sweat permeability of the VLD and 3D LD contributed to their superior wearable comfort capability (Fig. 4e). Due to the rapid transportation and evaporation of sweat, the VLD and 3D LD demonstrated better thermal comfortability during physical activity compared to the commercial ECG and PDMS patches (Fig. 4f and Extended Data Fig. 4). Infrared images revealed that after running for 15 minutes, the average skin temperature rose nearly 2.2 degrees under the commercial ECG electrode, while the VLD and 3D LD caused an average temperature rising of only 0.4 degrees (Fig. 4g).

Finally, we evaluated the effect of sweat accumulation on biosignal acquisition using three types of electrodes: commercial ECG electrodes, PDMS/Au serpentine-based dry electrodes, and 3D LD/Au serpentine electrodes (Fig. 4h). Although the integration of the Au mesh to the 3D LD marginally elevates the skin-electrode impedance, the 3D LD proffers superior adhesion and insulates against ambient interference (Supplementary Fig. 50). We affixed electrodes to a subject in a single-lead ECG configuration and tested the ECG signals of the subject before and

after 10- and 20-minutes exercise. All electrodes show high quality ECG data before exercise/sweating. However, after 10 minutes of running, the signals of the commercial ECG electrodes and those PDMS based ones became unstable with obviously decreased signal-to-noise ratio (SNR). The SNR of the commercial ECG and PDMS electrodes continued to deteriorate with longer duration of exercise/sweating, until became progressively difficult to read as more perspiration accumulated at the electrode-skin interface. In contrast, the 3D LD based electrodes featured great moisture permeability, that provided stable skin-electrode impedance and ECG signals both before and after sweating (Supplementary Fig. 51).

Integrated system-level sweat-permeable electronics

We demonstrate the universality and scalability of the 3D LD technology in two exemplary wearable electronic devices: a thin, soft ECG monitoring system and a textile-integrated weather station (Fig. 5a, b). The ECG monitor utilizes a detachable design through a magnetic coupling method. Component 1 is the vapor/sweat-releasing substrate, with a thickness of 0.65 mm, weight of 1.05g, bending stiffness of 55 kPa, and an elastic modulus of 106 kPa. Component 2 represents the flexible, waterproof circuitry layer, with a thickness of 2.1 mm and weight of 1.55 g including the battery (Supplementary Figs 52-54). The soft, stretchable, and wireless circuit was developed by the well-established skin electronics technologies^{11,42}, where a via connects the skin electrodes to the electronic connectors. (Fig. 5c and Supplementary Figs. 55, 56). The sophisticated circuit design ensures stable functionality under various deformations (Extended Data Fig. 5 and Supplementary Video 5). We affixed the device on the subject's chest, and the ECG data was wirelessly transmitted to a smartphone (Fig. 5d and Supplementary Video 6). We compared the motion artifacts between our device and the conventional cable-

1 connected type in daily activities and exercises including walking, jumping, and cycling
2 (Extended Data Figs. 6, 7 and Supplementary Video 7). Our device exhibits superior ECG
3 readouts under both conditions, owing to its low weight, conformable design, and strong
4 adhesion during the activities. The infrared images also exhibit lower thermal accumulation in
5 our device (Extended Data Fig. 8). To conduct long-term and user-friendly ECG monitoring,
6 we employed our ECG monitor for this purpose, as which exhibits exceptional moisture
7 permeability and conformability. Due to battery storage limitations, we designed a magnetic
8 charging cable to charge the device when needed (Supplementary Fig. 57). Throughout the
9 testing period, parameters such as heart rate (HR), QT interval (QT_o), and heart rate variability
10 (HRV) were successfully monitored (Fig. 5e). On a day (Wednesday) during the week when
11 the subject exercised, our device demonstrated superior capabilities in maintaining dry skin
12 conditions and producing stable signals during perspiration (Extended Data Fig. 9).
13 Furthermore, an increase in HR and a decrease in QT_o and HRV were observed as a result of
14 the physical activity (Fig. 5f). By increasing the layers of the circuit, the footprint of the circuit
15 can be minimized, and the device can be further integrated into a more compact format, and
16 therefore for multifunctional sensor, i.e. used for continuous electromyography (EMG)
17 monitoring (Extended Data Fig. 10). Another demonstration involves textile-integrated
18 moisture permeable electronics, as conventional textile electronics typically present a trade-off
19 between permeability and functionality or integration level. To address this issue, we developed
20 a battery-free, flexible, permeable, multifunctional, and wireless textile-integrated electronic
21 device, functioning as a T-shirt based weather station capable of monitoring temperature,
22 humidity, ultraviolet (UV) index, atmospheric pressure, and altitude (Fig. 5g and

Supplementary Fig. 58). Utilizing near-field communication (NFC) technology, weather data can be transmitted to a mobile device, as demonstrated during a hiking expedition (Fig. 5h, i). These integrated system-level sweat-permeable wearable electronic implementations indicate the adaptability and scalability of our 3D LD technology.

Conclusion

The 3D LD offers integrated system-level moisture permeability, surpassing human perspiration rates by three orders of magnitude, that enables seamless, continuous, and thermally comfortable healthcare monitoring. Comprehensive user studies show its superior performance in maintaining a stable and comfortable skin-device interface during sweating conditions. The ECG monitor based on 3D LD demonstrates reduced motion artifacts compared to conventional counterparts, yielding reliable signals over a weeklong duration. This positions 3D LD as a promising candidate for user-friendly, long-term healthcare monitoring wearables. Optimizing functional materials, device architecture, and manufacturing techniques is essential for enhancing its capabilities and achieving mass production. Incorporating a washable and recoverable adhesive back is expected to improve the device's reusability and cost-effectiveness.

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5

1 Figures

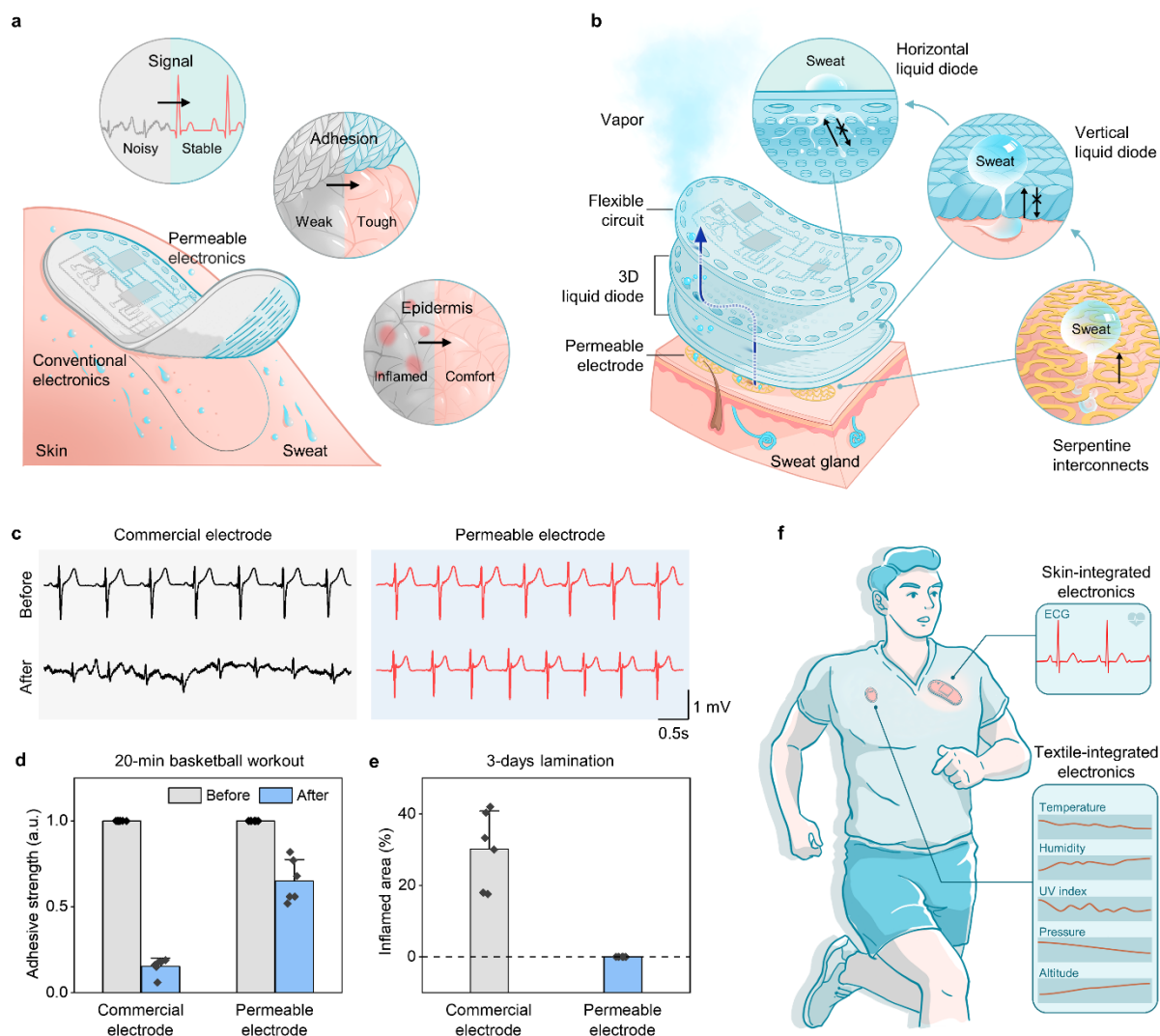


Fig. 1 Integrated system-level sweat-permeable wearable electronics based on 3D LD. a, Comparison between conventional flexible electronics (left) and permeable electronics (right). Permeable design allows for enhanced breathability and sweat wicking, improving signal stability, adhesion strength, and wearable comfort under sweating. **b,** Schematic of the integrated system-level sweat-permeable electronics, consisting of permeable electrodes, 3D LD for unidimensional sweat transport, and flexible circuit board. Blue arrow indicates the pathway of the sweat from the skin to the outlet. Exploded view illustrates the unidirectional sweat transport through the electrode, VLD, and HLD. **c,** ECG signals recorded by commercial and permeable electrodes before and after exercise. **d,** Adhesion strength evaluation of electrode-skin interface before and after a 20-minute basketball workout. Bar height, mean; error bars, s.d.; n = 6 independent samples. **e,** Skin irritation assessment of different electrodes after 3 days of wear. Bar height, mean; error bars, s.d.; n = 6 independent samples. **f,** Sweat-permeable skin-integrated and textile-based electronics for enhanced comfort and long-term monitoring.

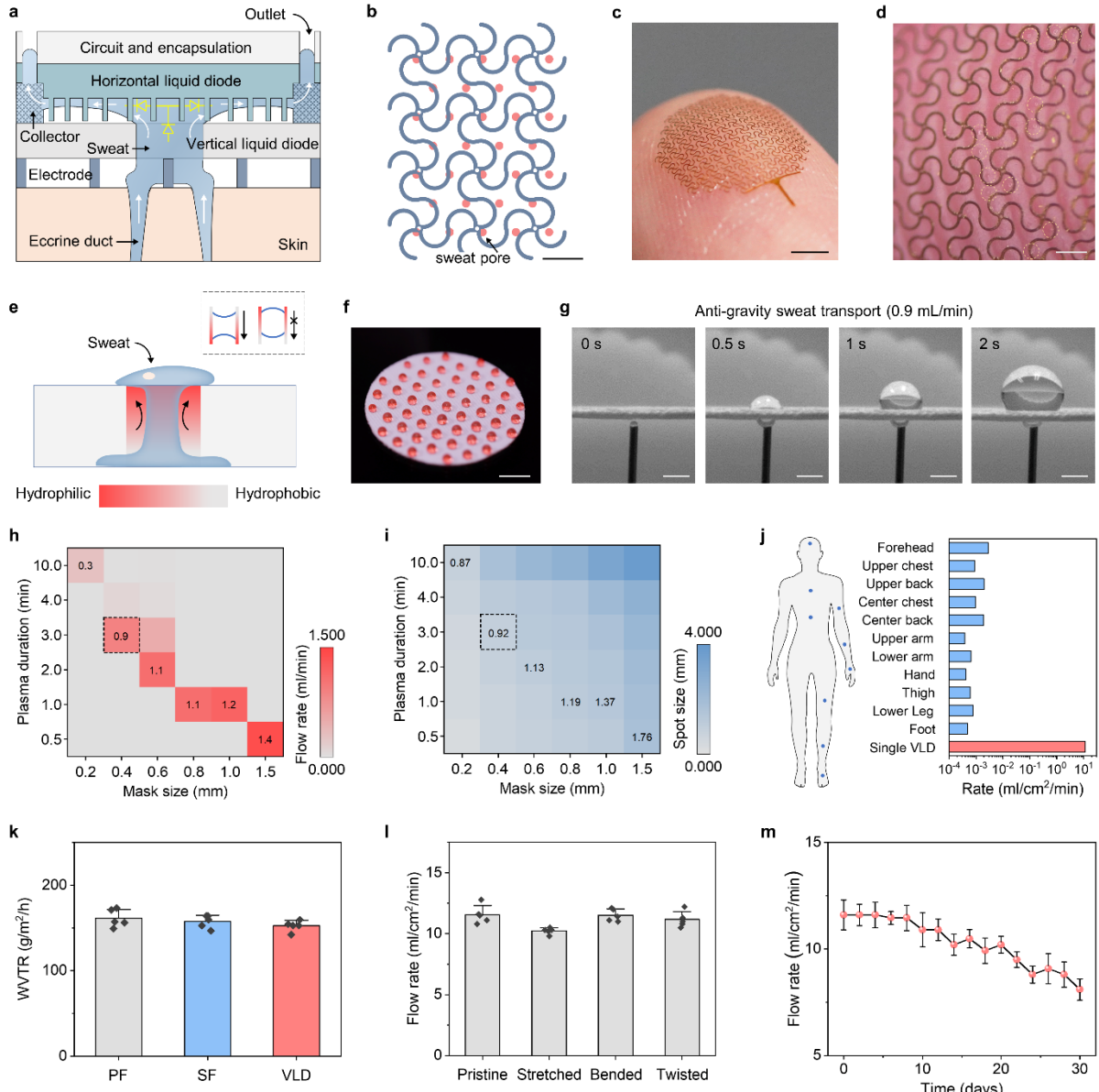
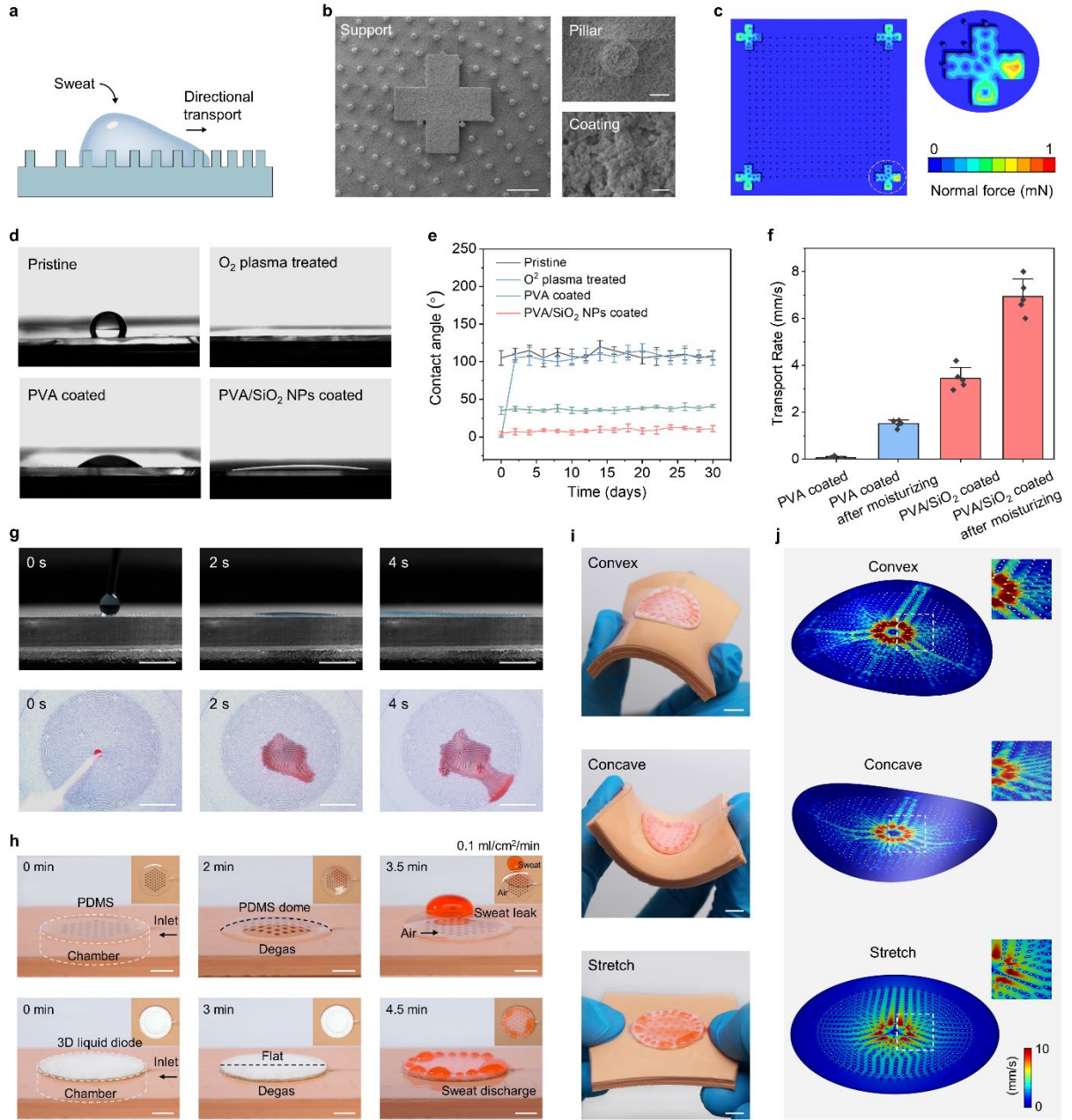


Fig. 2 Characterizations of the permeable electrode and VLD. **a**, Cross-sectional representation of 3D LD, demonstrating unidirectional sweat transport from the skin-device interface to the outlet. **b**, Design of serpentine interconnect to facilitate open channels above sweat pores. Scale bar, 0.5 mm. **c**, Permeable electrode maintains stable conformal contact on the fingertip under perspiration conditions. Scale bar, 2 mm. **d**, Magnified view illustrating sweat wicking within the open channels, with yellow circles highlighting sweat droplets. Scale bar, 0.5 mm. **e**, Mechanism of the unidirectional sweat transport in VLD. **f**, Photograph of VLD in the sweat wicking state. Scale bar, 5 mm. **g**, Anti-gravity sweat transport in a single channel of the VLD under a flow rate of 0.9 ml/min. Scale bar, 1 mm. **h,i**, Analysis of sweat transport rate (**h**) and spot size (**i**) of the single channel under various processing conditions. **j**, Sweat rate mapping of the human body during physical activity and the VLD. **k**, WVTR comparison between pristine fabric (PF), superhydrophobic fabric (SF), and VLD. Bar height, mean; error bars, s.d.; $n = 5$ independent samples. **l**, Sweat transport rate of the VLD under stretching, bending and twisting. Bar height, mean; error bars, s.d.; $n = 5$ independent samples. **m**, Long-

1 term stability assessment of the single channel in VLD. Points, mean; error bars, s.d.; n = 5
2 independent samples.

3

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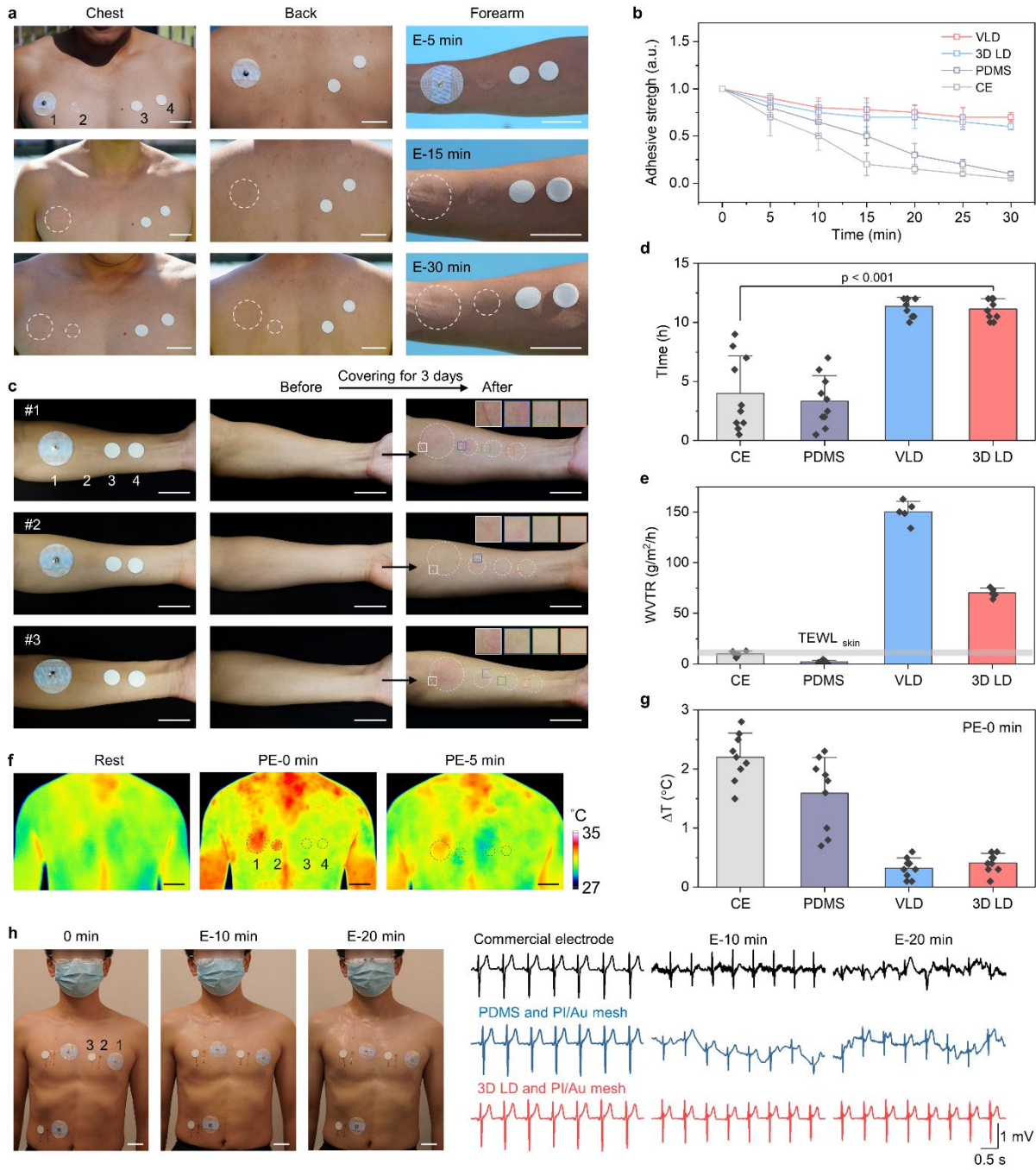
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Fig. 3 Characterizations of the HLD and 3D LD. **a**, Unidirectional sweat transport mechanism in HLD. **b**, SEM images of the supporting structure, pillars, and surface coating in HLD. Scale bar, 300 μm (support), 30 μm (pillar), and 2 μm (coating). Experiments were repeated three times with similar results. **c**, Computational modeling predictions of normal force distributions in HLD under stress conditions. **d**, Contact angle measurements of various PDMS surfaces: pristine PDMS, plasma-treated PDMS, PVA-coated PDMS, and PVA/SiO₂ nanoparticle-coated PDMS. **e**, Time-dependent contact angle measurements of PDMS surfaces. Points, mean; error bars, s.d.; $n = 3$ independent samples. **f**, Sweat transport rate in HLD with varying surface states. Bar height, mean; error bars, s.d.; $n = 5$ independent samples. **g**, Unidirectional sweat transport capabilities of HLD. Scale bar, 1.5 mm (cross-section view), 5

1 mm (top view). **h**, Comparison of PDMS membrane and 3D LD in terms of gas and sweat
2 permeability. Scale bar, 0.5 cm. **i**, Continuous sweat transport in 3D LD under bending and
3 stretching conditions. Scale bar, 1 cm. **j**, Fluid simulation of sweat transport in the deformed
4 3D LD.

5

1



2

Fig. 4 Evaluation of moisture management on wear comfort and device performance. **a**, Adhesion test was performed on commercial ECG electrodes (1), PDMS (2), VLD (3), and 3D LD (4) when attached to the skin during physical activity. Scale bar, 5 cm. **b**, Adhesion strength between patches and skin at different time intervals during exercise. Points, mean; error bars, s.d.; n = 3 independent tests. **c**, Skin irritation results for various patch types on the forearms of 3 volunteers. Insets display a closer view of the skin condition, magnified 3 times. Scale bar, 5 cm. **d**, Participants provided feedback on the wearing duration of different patches during a 12-hour test period. Bar height, mean; error bars, s.d.; n = 10 volunteers. Statistical significance

was assessed using two-sided unpaired t-tests. **e**, WVTR of the commercial ECG electrode, PDMS, VLD, and 3D LD. Bar height, mean; error bars, s.d.; n = 5 independent samples. **f**, Infrared images of various patches on skin before and after exercise. Scale bar, 5 cm. **g**, Temperature variations of the skin were compared while wearing patches against that of bare skin after exercise. Bar height, mean; error bars, s.d.; n = 9 independent spots. **h**, Images display commercial electrodes (1), PDMS/PI/Au mesh electrodes (2), and 3D LD/PI/Au mesh (3) on skin before and after exercise. Scale bar, 5 cm. **i**, ECG signals recorded from different electrodes before and after exercise.

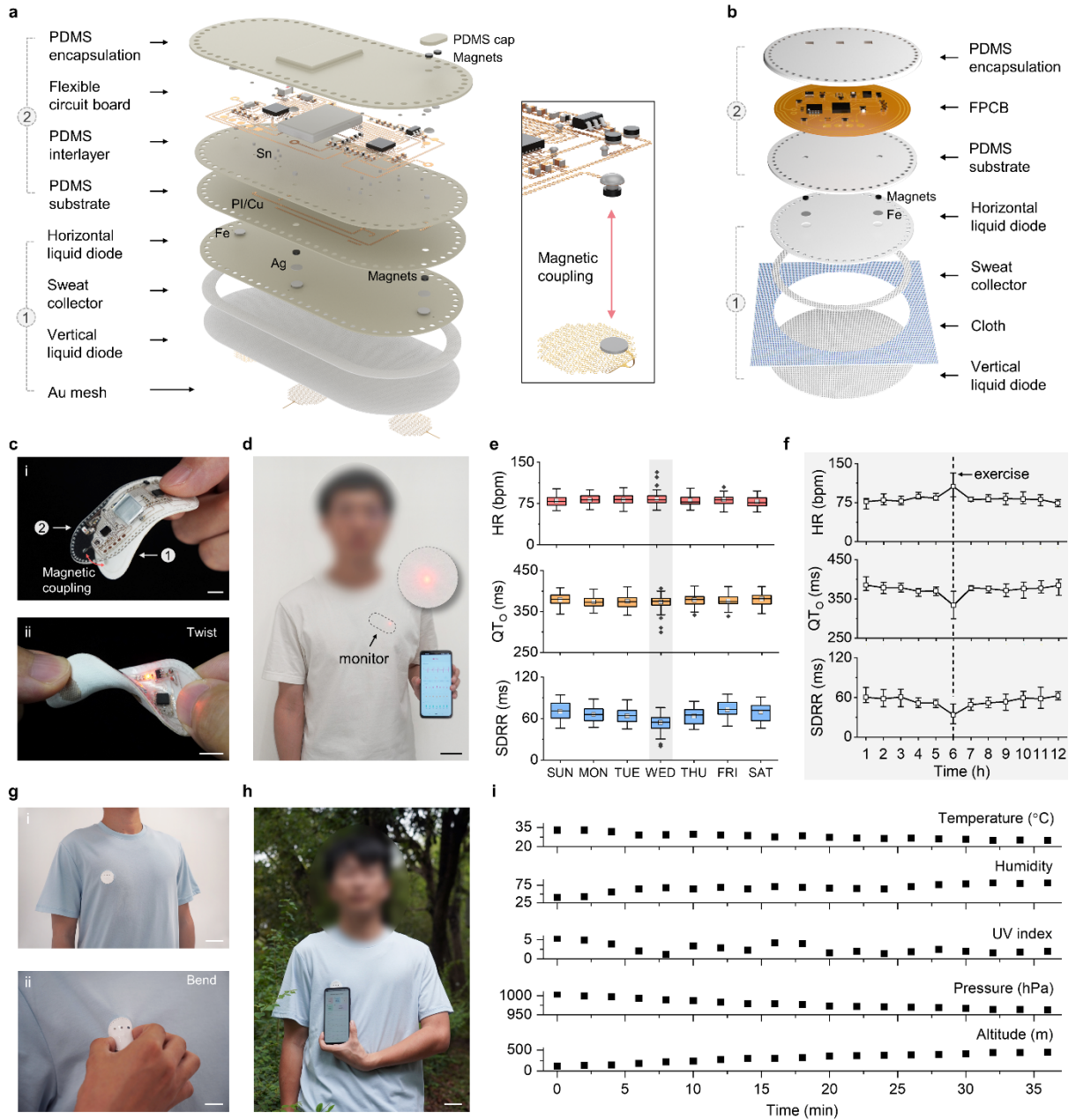


Fig. 5 Breathable and sweat-permeable skin-integrated and textile-based electronics. a,b, Exploded-view illustration of the skin-integrated (a) and textile-based (b) devices featuring a detachable design, highlighting key layers. Component 1 represents the vapor/sweat-discharging substrate, while component 2 refers the flexible circuitry layer. A magnified view elucidates the magnetic coupling mechanism between the two parts. **c,** Optical images of the permeable ECG monitor in a detached configuration (i) and a twisting state (ii). Scale bar, 1 cm. **d,** Photograph of the permeable ECG monitor on the upper chest, with signal transmission occurring wirelessly to a mobile phone. Scale bar, 5 cm. **e,** Daily ECG outcomes collected using the proposed device over a one-week period. HR refers to heart rate, QT_O donates QT interval, standard deviation of R-R intervals (SDRR) reflects the heart rate variability. Square, mean;

1 centre line, median; box limits, upper and lower quartiles; whiskers, $1.5\times$ interquartile range;
2 points, outliers; $n = 72$ independent spots. **f**, Alternations in HR, QT_O, and SDRR before and
3 after physical activity. Points, mean; error bars, min-max; $n = 6$ independent spots. **g**,
4 Photograph of the permeable textile-integrated weather station. Scale bar, 5 cm (i), 2 cm (ii). **h**,
5 optical image of a subject wearing the T-shirt weather station while hiking. Scale bar, 5 cm. **i**,
6 Real-time meteorological data captured during the hiking excursion.

7

Methods

Fabrication of 3D LD. The fabrication process is divided into the fabrication of the VLD⁴³, HLD, and assembly processes. Fabrication of the VLD began with laser structuring of the hydrophilic polyester fabrics (LPKF ProtoLaser U4) followed by cleaning with acetone, isopropyl alcohol, and deionized water and then drying at 50°C for 2 hours. 4 ml of 1H, 1H, 2H, 2H-perfluorooctyltriethoxysilane (PFOTES), 10 g of P25 TiO₂ nanoparticles, and 1 ml of deionized (DI) water were dispersed into 200 ml of ethanol via stirring for 30 min to form a suspension. The tailored polyester (PET) fabrics (150 μ m thick) were immersed into the suspension stirred for 2 hours and dried at 50°C for 2 hours to yield the superhydrophobic fabric. Single surface dip-coating in the water-based acrylic pressure-sensitive adhesives (PSA) formed the permeable adhesive network (Supplementary Fig. 10), and the optimal coating repetition is 5 times. Silicone PET release film served as the protective layer of the back adhesive and laser-structured polyimide (PI) tapes (50 μ m) served as the masks of the fabrics for selective oxygen plasma treatment (200W, 200 mTorr, O₂ flow rate of 50 sccm, PlasmaTherm 790).

The HLD system fabrication was based on photolithography and replica molding of polydimethylsiloxane (PDMS; 10:1 of base to curing agent, Sylgard-184, Dow Corning, USA). The layout was designed with computer graphics software (AutoCAD, Autodesk, USA) and transferred by chrome masks. Fabrication of the HLD started from spin casting a 100- μ m-thick film of negative photoresist (SU8 2050) on an O₂ plasma treated 4" silicon wafer followed by soft baking at 65°C for 3 min and 95°C for 20 min on a hot plate. Exposing the wafer to ultraviolet light through a mask patterned the photoresist. The wafer was post-exposure baked at 65°C for 5 min and 95°C for 10 min followed by immersing in the developer solution for 10 min. Laminating laser-patterned PET films to the wafer served as the mold. Molds were treated with Octadecyltrichlorosilane (OTS) by vapor silanization to facilitate the release of the PDMS micropillars. Pouring PDMS (10:1 of base to curing agent, Sylgard 184, Dow Corning) into the mold, followed by degassing under vacuum for 1 hour and then baking at 80°C for 30 min. After release from the mold, a mechanical punch aligned by the laser-structured PET mask defined the outlet holes. 0.1 g of hydrophilic fumed silica and 0.05 g of sodium dodecyl sulfate (SDS) were dispersed into 10 ml of 1wt% polyvinyl alcohol (PVA) solution to form the hydrophilic coating after sonication for 10 min. Spray coating the hydrophilic PVA composite solution onto the plasma treated PDMS microfluidic film performed on a hot plate at 110°C, yielded the long-term superhydrophilic layer.

Laser-structured polyester fabric (150 μ m thick) served as the sweat collector after cleaning with acetone, isopropyl alcohol, and deionized water and then drying at 50°C for 2 hours. VLD, sweat collector, and HLD were bonded via the sil-poxy silicone adhesive.

Fabrication of the PI/Au open-mesh electrode. Spin casting PDMS (500 rpm, 30s) on a temporary glass sheet followed by baking at 100°C for 10 min. 20- μ m thick PI film was laminated on the PDMS and patterned into open mesh architectures (trace width of 50 μ m) by laser cutting (LPKF ProtoLaser U4). Water-soluble tape (WST) was used to pick up the completed patterns from the glass substrate. Ti/Au (5/50 nm) thin film was deposited on the patterned PI films by electron beam evaporation. Removing WST by immersing it into DI water formed the self-supported open-mesh electrode.

Fabrication of the wireless ECG monitor. The ECG monitor includes two parts, the vapor/sweat-discharging substrate and flexible circuit components. Two-step casting PDMS into the mold and assembling laser-patterned iron disks (0.25 mm diameter and 0.15 mm thickness; LPKF ProtoLaser U4) formed the HLD. The thickness, length, width, and weight of the VLD are 150 μm , 65 mm, 30 mm, and 0.3 g respectively, while the HLD measures 500 μm in thickness, 65 mm in length, 30 mm in width, and weighs 0.6 g. After assembling as the 3D LD, three through holes (0.5 mm diameter) in PDMS were defined by a mechanical punch. PI/Au open mesh electrodes passed through the fabric and through holes and were bonded with iron discs by the silver paste followed by baking at 80°C for 30 min. Restituting the punched-out PDMS pillars to the through holes and bonding by the sil-poxy silicone adhesive. PET release film was attached to the bottom of the substrate to protect the adhesive.

The flexible circuit began with laminating the laser-structured PET sheet to the silicon wafer as the mold, followed by placing the magnets (1.5 mm diameter and 0.5 mm thickness) and pouring PDMS into the mold and curing at 80°C for 10 min. Molds were treated with Octadecyltrichlorosilane (OTS) by vapor silanization to facilitate release of the PDMS film. Spin casting PDMS (500 rpm, 30s) on a temporary glass sheet followed by curing at 100°C for 10 min. Cu/PI sheet (Cu 18 μm , PI 12.5 μm) was laminated on the prepared PDMS/glass substrate and then structured into designed pattern by laser cutting (LPKF ProtoLaser U4). The pattern was picked up by WST and Ti/SiO₂ (5/50 nm) thin film was deposited by electron beam evaporation. The first layer of patterned PI/Cu interconnection was directly laminated on the molded PDMS substrate after activating by ultraviolet ozone for 5 min and heated at 80°C for 10 min. Removing the WST by DI water. Placing laser-patterned PET stamps (0.6 mm diameter and 1 mm thickness) with thermal sensitive adhesive on the Cu pads. Repeated pouring of PDMS into the mold and curing at 40°C for 12 h followed by removing the PET stamps at 60°C formed the vertical interconnect accesses (VIA) through the PDMS. Transfer printing the second layer of PI/Cu pattern onto the PDMS in the same way and removing the WST by DI water. Soldering the electrical components (microcontroller, resistors, capacitors, blue tooth, light-emitting diode, et al.), magnets, battery, and Cu interconnected pads with the low melting point solder paste (Supplementary Fig. 59). Sealing the circuit by the PDMS (~ 100 μm thick) and curing at 80°C for 10 min followed by removing from the mold and laminating the PET release film. Molding a PDMS cap to protect the magnets from ambient sweat. For operation, magnets in the flexible circuit and iron discs in the substrate were automatically aligned and mechanically coupled, building the electrical conduction from the electrode to the circuit.

Fabrication of the magnetic charging cable. The top and bottom covers were fabricated by photocuring 3D printing. Assembling the USB cable with the top cover and soldering the Cu wires on the magnets (1.5 mm diameter, 1 mm thickness) with the low melting point solder paste, followed by installing the bottom cover.

Fabrication of the textile integrated wireless weather station. The wireless weather station includes the textile integrated vapor/sweat-discharging substrate and flexible circuit. Fabrication of the vapor/sweat-discharging substrate was the same as that of the ECG monitor without creating the back adhesive and electrode. The diameter of the VLD (46 mm) was larger than that of the HLD (36 mm). Laser structuring clothing with a hole (36 mm diameter)

1 followed by bonding the 3D LD to the clothing via the sil-poxy silicone adhesive. A customized
2 flexible circuit board (FPCB) was prepared, and then electrical components and magnets (1.5
3 mm diameter and 0.5 mm thickness) were soldered (Supplementary Fig. 60). Placing the FPCB
4 into the 3D printed mold and pouring the PDMS with white pigments (silc pig, Smooth On,
5 Inc.), followed by degassing and curing at 80°C for 10 min. PI shadow masks were used to
6 insulate the sensors (temperature, humidity, UV, and pressure) from the PDMS.

7
8 **Contact angle measurement.** Contact angles (CAs) were measured using an optical CA meter.
9 The PDMS films were mounted on the glass sheets and the fabric strips were fixed to the 3D-
10 printed frame, and liquid drops with a volume of 10 μ L were used to check the CA and transport
11 direction. A linear motor is utilized to measure the CA of the fabrics at various strains.

12
13 **Anti-gravity sweat transport test of VLD.** VLDs were fixed on the 3D-printed frame and a
14 needle (0.2 mm inner diameter) was placed beneath the selectively plasma treated spot with a
15 gap of 1 mm. Continuous artificial sweat flow with a controlled rate was generated from a
16 syringe pump. A camera was placed front of the fabric to record the sweat transport behavior.

17
18 **Sweat transport test of HLD.** Artificial sweat droplets (20 μ L) with red dye were dripped onto
19 the HLD with different surface properties. A camera was set above to capture the sweat transport
20 process, sweat transport rates before and after moisturizing were calculated by dividing the
21 transport distance by the time. For the cross-section view, the HLD was cut radially, and the
22 camera was set in front. The real-time morphology variation of the micropillars at sweat flow
23 was captured via the inverted microscope (ECLIPSE Ts2R, Nikon).

24
25 **Sweat transport test of 3D LD.** Spin casting PDMS with flesh pigments (silc pig, Smooth On,
26 Inc.) on a temporary glass sheet followed by baking at 80 °C for 10 min. Laser structuring the
27 holes (1 mm diameter and interval) through the PDMS. Pouring the PDMS with flesh pigments
28 into the 3D-printed mold and baking at 80°C for 30 min formed the substrate with an open
29 chamber. Assembling the perforated PDMS film, silicon tube (0.3 mm diameter), and PDMS
30 substrate via sil-poxy silicone adhesive simulated the sweat glands. Covering the 3D LD on the
31 imitative sweat glands and continuous artificial sweat flow with a controlled rate was generated
32 from a syringe pump. Two cameras were placed in front and above the device to record the
33 deformation of the 3D LD and the sweat flow process. For sweat wicking test, a drop of artificial
34 sweat droplets (20 μ L) with red dye was dripped on the volunteer's forearm. Covering the 3D
35 LD on the droplet for 4 seconds and then removing it. A camera was set above to capture the
36 sweat flow in the 3D LD.

37
38 **WVTR measurement.** The water vapor transmission rate test was conducted at 17°C with the
39 humidity of 67%. 30 ml of DI water was filled into a 50 ml bottle (15 mm diameter) and
40 completely sealed by the samples. The water mass lost was tested after 12 hours. The WVTR
41 was calculated by dividing the loss mass by the dimension of the opening area and the time
42 duration.

43
44 **Stability test.** Stability test was conducted on the VLD and HLD over a period of 30 days. The

skin surface area of a person is estimated at 1.8 m², and the sensible sweat loss volume is approximately 10 L at long-term physical work⁴¹. Thus, the daily sweat usage in the test is set to 0.6 ml/cm². The VLD was securely fastened onto a 3D-printed frame and a needle (0.2 mm inner diameter) was placed beneath the selectively plasma treated spot with a gap of 1 mm. Continuous artificial sweat flow with a controlled rate was generated from a syringe pump. The working area of the HLD in Fig. 3g is 2.2 cm², hence the daily volume of sweat needed is estimated at about 1.3 ml. Continuous artificial sweat flow (0.1 ml/min) was generated from a syringe pump while capturing videos from above.

Resistance stability of the open-mesh electrode. Spin casting PDMS (500 rpm, 30s) on a temporary OTS-treated glass sheet followed by baking at 100°C for 10 min. The PI/Au open-mesh electrode was picked up by WST and Ti/SiO₂ (5/50 nm) thin film was deposited by electron beam evaporation. The PI/Au open-mesh electrode was directly laminated on the PDMS substrate after activating by ultraviolet ozone for 5 min and heated at 80°C for 10 min. Fixing the PDMS substrate on a biaxial stretching system and measuring the resistance of the electrode at various strains from 0% to 30%.

Cell cytotoxic test. The mouse embryo fibroblast cell line, NIH/3T3 (ATCC, US), served as the model for evaluating in vitro cytotoxicity of the VLD and 3D LD. These cells (1×10⁴ cells/mL) were cultured in Dulbecco's Modified Eagle's Medium (DMEM, Hyclone) supplemented with 10% fetal bovine serum (FBS) and 1% penicillin/streptomycin (PS, Procell) at 37°C, in a humidified incubator with 5% CO₂. Specimens were cut into 1 cm diameter disks, soaked in culture medium overnight, and seeded with NIH/3T3 cells in a 24-well plate. Cells were cultured for 24, 48, and 72 hours. After removing the medium, a phosphate buffer saline (PBS) solution containing 2 μM calcein-AM and 8 μM EthD-1 was added, and the samples were incubated for 30 minutes before examination with a fluorescent microscope. For cell proliferation evaluation, a CCK-8 assay (Beyotime Biotechnology) was conducted as per the manufacturer's instructions. Briefly, CCK-8 solution was added to each well at 10% of the culture medium's volume, followed by a 4-hour incubation. The absorbance at 450 nm, indicative of cell metabolic activity, was measured using a microplate reader.

Human experiment. Volunteer experiments were conducted to assess the impact of sweat on both device performance and the epidermis, including adhesion test, biocompatibility test, skin temperature evaluation, and ECG signal collection. Four samples (commercial ECG electrode, PDMS, VLD, and 3D LD) were attached to the chest, upper back, and right forearm for the adhesion test. The volunteer played basketball outdoors (28°C, 72% humidity) for 30 min and pictures of the samples were captured at 5, 15, and 30 min, respectively. To assess the adhesion strength between various patches and the skin on the back, we performed a 90-degree peel test utilizing a customized test system (Supplementary Fig. 48). This system consists of a clip connected to a highly sensitive force gauge (resolution, 0.001 N) via a 3D-printed connector. The stretching speed was regulated at nearly 50 mm/s, and the adhesion force was denoted by the peak value measured before the complete detachment of the patches from the skin. First, we evaluated the adhesion strength of each patch type against dry skin. Then, six sets of samples (commercial ECG electrodes, PDMS, VLD, and 3D LD) were affixed to a volunteer's back.

Following every 5-minute interval of a basketball exercise session, the patches were peeled off, and the corresponding peeling forces were documented. The skin irritation test was carried out on human skin. Four samples (commercial ECG electrode, PDMS, VLD, and 3D LD) were attached to the forearms of the volunteers. Patches were removed after being worn for 3 days, and skin was captured before and after the test. Skin reddening percentage was further quantitatively determined using ImageJ software. Ten volunteers participated in the wear comfort test and the test duration is 12 h. Four samples were attached to the forearms and the volunteers removed the patches if felt any discomfort, then record the wearing duration of each sample. The skin temperature evolution under the patches after exercise was performed with a thermal imaging camera (FLK-TIS60, Fluke). Four patches (commercial ECG electrode, PDMS, VLD, and 3D LD) were attached to the back of the volunteer, removing the patches after 15 min running and thermal imaging. The ECG test of volunteer after exercise is performed with commercial data acquisition system (PowerLab). Patches (commercial electrode, PDMS/PI/Au mesh, and 3D LD/PI/Au mesh) were attached to the chest and abdomen. Recording the ECG signals of the volunteer at rest and after running for 10 and 20 min. For long-term ECG monitoring, the sweat-permeable ECG monitor was attached to the left chest of the volunteer. ECG signals were recorded from 9 am to 9 pm considering the working period of the monitor is 5 h with a fully charged battery. A power bank with magnetic charging cable is used to charge the battery after the monitor working for 4 h. The ECG curves were analyzed by Origin and MATLAB. For wearable weather station operation, the volunteer wore the T-shirt and hiked on the Beacon Mountain trail. Recording the weather parameters every 2 min. All human experiments were conducted by protocols approved by the Human Subjects Ethics Subcommittee of Research Committee, City University of Hong Kong, Hong Kong, China, and guidelines were followed. All participants' informed consent was obtained before inclusion in this study.

Mechanical simulation. To numerically analyze the mechanical response and optimize the design parameters of various parts of our electronic, finite element analysis (FEA) was conducted. The 3D models were first constructed by *SOLIDWORKS 2022* and then meshed by *MSC Apex 2022*. All objects were meshed into linear hex elements. The mesh size was one-fifth of the width of the electrode, integrated circuit and the silicone support models, ensuring the convergency and accuracy. The mesh files were imported to *Marc Mentat 2022* for finite element model construction. Fiction was neglected in the simulation. The elastic modulus (E) and Poisson's ratio (ν) applied in the simulation were $E_{Cu} = 119$ GPa and $\nu_{Cu} = 0.34$ for copper; $E_{PI} = 0.8$ MPa and $\nu_{PI} = 0.5$ for PI; and $E_{PDMS} = 145$ kPa and $\nu_{PDMS} = 0.5$ for PDMS; $E_{Au} = 77.2$ GPa and $\nu_{Au} = 0.42$.

Fluid simulation. FEA commercial software COMSOL was employed to verify and optimize the design and performance of the 3D LD. To optimize the structure stability of the pillars under liquid flow, we created a 2×2 pillar array located in the bottom center of a cube with a side length of 1 mm. The diameter of the pillar is 50 μm , and height (H) and the interval between adjacent pillars (L) were varied. The density, Youngs Modulus, and Poisson's ratio were 970 kg m^{-3} and 970 kPa, and 0.49, respectively. The fluid had an incompressible laminar flow. Its density and viscosity were 1,000 kg m^{-3} and 1.01×10^{-3} Pa·s, respectively. The inlet velocity

was set as $V_{inlet} = 15 \cdot \sin(2\pi t)$ mm/s, and the outlet pressure was given as 0 Pa. All simulations used incompressible laminar flow model based on Navier-Stokes equation. The displacement along the pillar's centerline was analyzed. Three-dimensional models were constructed to simulate the sweat-discharging process of the 3D LD. The arrangement of the sweat-discharging patch array was imported from its AutoCAD file and was used to generate a layered structure consisting of a top layer of PDMS (0.5 mm thickness), a micropillar array (0.1 mm height), a bottom layer textile (0.15 mm thickness) with an inlet (0.9 mm diameter). The diameter of the patch used for the simulation was 10 mm. Both models were set to have an inlet flow rate and outlet pressure of 0.5 mm/s and 0 kPa, respectively. The simulation was set to consider the laminar two-phase flow of sweat (water) and its surrounding air (gas). To reveal the dynamic characteristics of the sweat-discharging process, we used the phase field module for time-lapsed computation. The temperature, reference pressure level and surface tension between the liquid-gas phase were 293.15 K, 1 atm and 72.8 mN/m respectively. Gravity of each phase was considered in the flow field simulation and contact angle between liquid phase and PDMS wall was 10° .

Conflicts of Interest

The authors declare no conflict of interest.

Data availability

All data supporting the results of this study are available within the paper and its Supplementary Information. Source data are provided with this paper.

Code availability

Custom codes for raw ECG signal collection are available at <https://doi.org/10.5281/zenodo.10575185>.

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Author contributions

X.Y. and B.Z. conceived the idea and designed the projects. B.Z. designed the whole system and conducted overall experiments. B.Z. and J.L. designed the liquid diode. J.Z. designed the circuits. L.C. and B.Z. conducted the mechanical modeling. Q.Z. and B.Z. carried out the cytotoxicity test. J.L., Z.M., and B.Z. conducted the fluid simulation. F.C. and X.H. performed the materials characterization. Y.Y., C.K.Y., Jian L., P.W., S.J., H.L., D.L., Y.L., K.Y., R.S., and Z.C. assisted in the fabrication and testing experiments. B.Z., L.C., Y.H., G.Z., J.Z., J.L., and Y.G. organized the user study. B.L.K., W.Y., F.W., Z.Z., and Z.W. evaluated the

- 1 experiments and contributed valuable ideas. B.Z. and X.Y. wrote the manuscript. All authors
- 2 discussed the results and commenting on the manuscript.

Extended Data Figure Legends

Extended Data Fig. 1 Contact angle of the selective plasma-treated fabric under various treatment conditions. **a**, Schematic illustration of the fabric samples for selective plasma treatment. **b**, Contact angle of the channels with mask size from 0.2 to 1.5 mm on both top (T) and back (B) sides of the superhydrophobic fabric after plasma treatment for 0.5 to 10 min.

Extended Data Fig. 2 Fluid simulation of the 3D LD. **a**, Velocity magnitude. **b**, Streamlines. **c**, Pressure contours. **d**, Spatiotemporal variation in the liquid volume fraction.

Extended Data Fig. 3 Cytotoxicity test of the VLD and 3D LD. **a**, LIVE/DEAD staining (green: live cell; red: dead cell). Scale bar, 100 μ m. **b,c** Cell viability (b) and optical density (OD) values (c) obtained in the in vitro cytotoxicity assay. Bar height, mean; error bars, s.d.; n = 3 independent samples.

Extended Data Fig. 4 Evaluation of skin temperature beneath various patches including commercial ECG electrodes (1), PDMS (2), VLD (3), and 3D LD (4). **a**, Sequential arrangement of patches 1, 2, 3, 4 from left to right. Scale bar, 5 cm. **b** Outline of the test procedure. **c, d**, Photographic and infrared images of skin pre- and post-exercise with different patches applied. Scale bar, 5 cm.

Extended Data Fig. 5 Optical images (a) and FEA modeling (b) of the permeable ECG monitor under stretching, bending, and twisting. Scale Bar. 1 cm.

Extended Data Fig. 6 Comparison of motion artifacts between the conventional cable-connected type and our wearable device. **a-d**, Raw ECG data (a,b) and post-processed data (c,d) from the commercial electrode connected with DAQ and our device under resting, walking, and jumping conditions. ECG data from DAQ was lost under walking and jumping conditions, whereas our device maintained stable ECG signal readout under body motion.

Extended Data Fig. 7 Continuous ECG recording during a 30-min cycling workout. **a**, ECG signals from the cable-connected DAQ are difficult to discern during the cycling process. **b** Continuous ECG signals from our wearable and permeable device. The heart rate (HR) and QT interval (QT₀) are calculated simultaneously and continuously. **c**, Enlarged view of the ECG signal, HR, and QT₀ during the cycling and rest periods.

Extended Data Fig. 8 Comparative assessment of the thermal comfort provided by commercial ECG electrodes and our permeable ECG monitor. Three volunteers are asked to wear both the commercial electrodes and our device while continuously cycling for 30 minutes. Infrared images of the skin are captured immediately after patch removal and following a 5-minute rest. The maximum temperature beneath the commercial electrodes was nearly 1.9 degrees higher than that beneath our device. Scale bar, 5 cm.

Extended Data Fig. 9 ECG monitoring by the permeable patch before and after running.
a,b Optical images of the patches attached to the chest before (a) and after exercise (b). Scale bar, 5 cm. **c**, Optical image depicting the condition of skin beneath the device. Scale bar, 1 cm. **d**, Optical image demonstrating the sweat discharging channels of the device. Scale bar, 1 cm. **e,f**, ECG signals acquired before (e) and after exercise (f).

Extended Data Fig. 10 Four-layer integrated circuit. **a**, Exploded schematics illustrating the layer-by-layer structure. **b**, Photographic image of the integrated circuit. Scale bar, 1 cm. **c,d**, Images of the permeable device affixed to the chest (c) and forearm (d) for ECG (c) and electromyography (EMG; d) signal acquisition, alongside the recorded ECG (c) and EMG (d) traces. Scale bar, 6 cm.